GestureVoice-5G: Real Time Sign-to-Speech Translation

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ABSTRACT

People with speech and hearing disabilities often struggle to communicate in hospitals, classrooms and public services, where a sign language interpreter is not always available. This work presents **GestureVoice-5G**, a real-time system that converts sign gestures into spoken output. An Arduino UNO Q extracts 42 hand landmarks per frame with MediaPipe and sends them, never video, over a 5G CPE local network to a multi-access edge computing (MEC) server, where a lightweight GRU classifier recognises 27 signs from 45-frame windows. On held-out validation clips the classifier reached **96.8% accuracy** (macro F1 95.8%). The mean round trip, including inference, was **34 ms** to the MEC server over the CPE's local network, against 166 ms to a laptop over the campus network. Landmarks are about 50× smaller than a compressed video frame, and no video leaves the capture device.

INTRODUCTION

- Most people cannot understand sign language, so deaf and speech-impaired people often depend on interpreters.
- Gesture recognition has matured, but a usable system must be fast: delays of a few hundred milliseconds break a conversation.
- **GestureVoice-5G** converts hand gestures into speech in real time.
- Edge computing (MEC) over a private 5G network aims to keep latency low and links reliable.

OBJECTIVES

- Convert sign gestures into speech in real time.
- Keep the network round trip low by running inference on a 5G edge (MEC) server.
- Recognise a 27-sign vocabulary accurately from hand landmarks.

SYSTEM MODEL / PROBLEM FORMULATION

- **System:** IP camera → Arduino UNO Q → (Wi-Fi) 5G CPE → (Ethernet) MEC server.
- **Input / output:** a 45-frame window of hand landmarks X; one of 27 signs ŷ.
- **Delay:** To minimise the round trip $T_{\mathrm{rt}} = T_{\mathrm{up}} + T_{\mathrm{inf}} + T_{\mathrm{down}}$ (mean 34 ms); UNO Q processing adds $T_{\mathrm{cap}} \approx 400$ ms per frame.
- **Notation:** T_{up}, T_{down} = upload, return; T_{inf} = inference; T_{cap} = UNO Q capture and tracking.

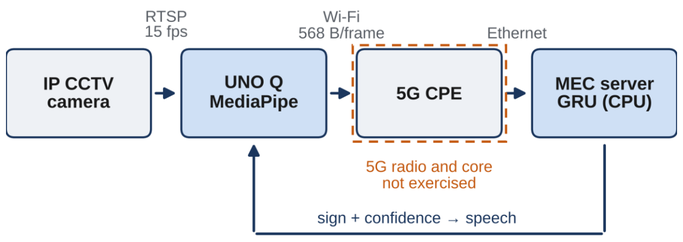


Figure 1: System as tested. The 5G radio and core were not exercised.

PROPOSED METHOD / METHODOLOGY

- **Capture:** the UNO Q decodes the 15 fps CCTV stream and extracts 42 hand landmarks per frame with MediaPipe.
- **Streaming:** only landmarks (568 bytes per frame, no video) go over TCP/WebSocket to the MEC server.
- **Classification:** a GRU classifier (PyTorch, CPU) labels 45-frame windows into 27 signs and replies with votes and a confidence threshold.
- **Speech:** the sign is returned and spoken by text-to-speech.

EXPERIMENTAL / SIMULATION SETUP

- **Hardware:** CCTV camera, Arduino UNO Q, 5G CPE, Dell MEC server (CPU).
- **Network:** UNO Q → CPE over Wi-Fi, CPE → MEC server over Ethernet. The 5G core was down, so the 5G radio was not used.
- **Baseline:** campus network → MacBook laptop.
- **Data:** 2,089 clips, 27 signs; 80/20 split stratified by sign; 411 held-out clips.
- **Metrics:** accuracy, F1, mean round trip (343 and 348 predictions), UNO Q frame rate.

RESULTS AND DISCUSSION

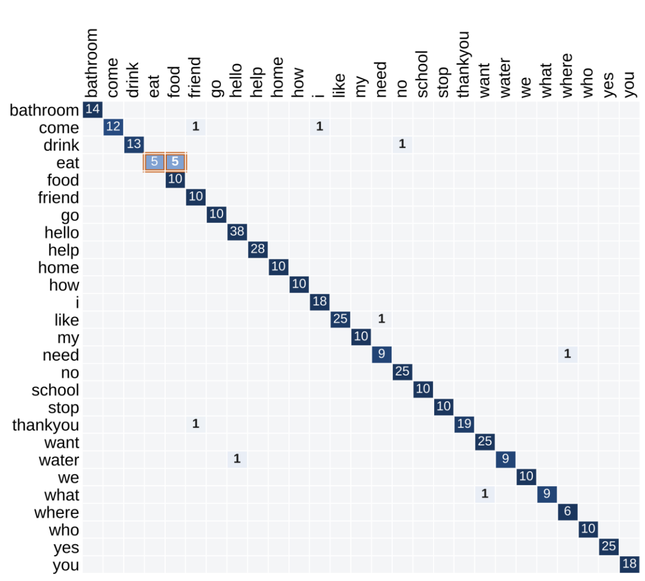


Figure 2: Confusion matrix, 411 held-out clips. “eat” → “food” in 5 of 10 clips; all other errors occur once.

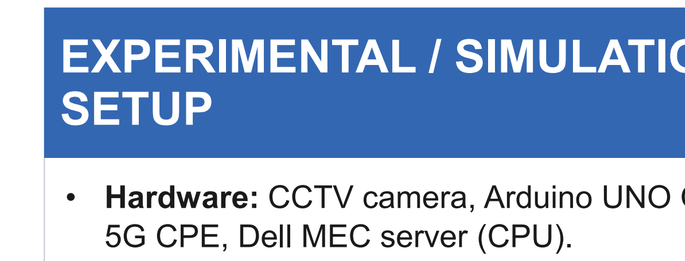


Figure 3: Mean round trip, two setups (different servers and days). Campus path with live camera: 364 ms (n = 66).

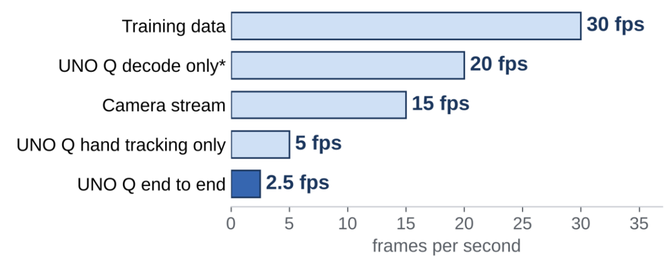


Figure 4: Frame rate per stage. *Decoder catching up on buffered frames. 45 frames = 18 s at 2.5 fps, 1.5 s at 30 fps.

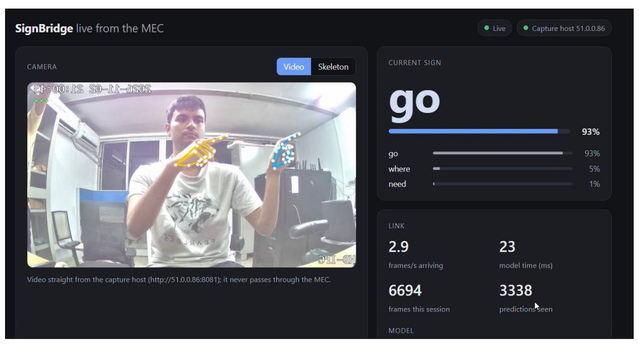


Figure 5: Live demo page: sign, confidence and caption (laptop run).

- **Latency:** inference took 6–27 ms. Mean round trip 34 ms (CPE local network → MEC) vs 166 ms (campus → laptop); different servers and days.
- **Accuracy:** 96.8% on held-out clips from the training sessions and signers; new signers will likely score lower.
- **Trade-off:** landmarks only, so no video leaves the device. But the UNO Q ran at ~2.5 fps vs 30 fps in training, which hurts fast signs.
- **Limits:** 5G radio and core not tested; end-to-end delay not measured.

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CONCLUSION

- Built and tested: UNO Q hand landmarks → MEC server → GRU classifier → speech.
- 96.8% accuracy on 27 signs; mean round trip 34 ms over the CPE local network.
- Next: test over the 5G radio and UPF, measure end-to-end delay, test new signers.

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